

PyCharm & Jupyter Notebook

Installation and Virtual Environment Setup Guide

1 Installing PyCharm for Data Science

PyCharm is a powerful IDE by JetBrains. For native, fully-featured Jupyter Notebook support, **PyCharm Professional** is highly recommended, though community plugins exist for the Community Edition.

1. Navigate to the official JetBrains website: <https://www.jetbrains.com/pycharm/download/>
2. Download the installer for your operating system (Windows, macOS, or Linux).
3. Run the installer and follow the on-screen instructions.
4. Launch PyCharm and select **New Project**. Choose a directory for your workspace (e.g., `capstone-lab`).

2 Creating a Virtual Environment (venv)

A virtual environment isolates your project's dependencies from your global Python installation.

1. install Python 3.x from (python.org), if not already present. The installation instructions are available at: [python.org]
2. Open the **Terminal** inside PyCharm (usually located at the bottom of the window, or accessible via `Alt + F12` / `Option + F12`).
3. Ensure you are in your project's root directory.
4. Run the following command to create a virtual environment named `venv`:

Terminal - Create venv

```
python -m venv venv
```

4. Activate the virtual environment. The command depends on your operating system:
For Windows (Command Prompt / PowerShell):

```
.\venv\Scripts\activate
```

For macOS and Linux:

```
source venv/bin/activate
```

Note: You will know the environment is active when you see `(venv)` prepended to your terminal prompt.

3 Installing Dependencies from requirements.txt

With your virtual environment active, you must install the required libraries (including Jupyter itself) using your `requirements.txt` file.

1. Ensure your `requirements.txt` file is placed in the root of your project directory.
2. Upgrade pip to the latest version (best practice):

```
python -m pip install --upgrade pip
```

3. Install the requirements:

```
pip install -r requirements.txt
```

Crucial Requirement

To run notebooks, your `requirements.txt` **must** include the `jupyter` or `notebook` package.

4 Configuring PyCharm to Execute .ipynb Files

Now that the environment is set up and dependencies are installed, you need to tell PyCharm to use this specific `venv` to run your notebooks.

4.1 Step 1: Set the Python Interpreter

1. Go to **File** → **Settings** (Windows) or **PyCharm** → **Settings** (macOS).
2. Navigate to **Project: [Your Project Name]** → **Python Interpreter**.
3. Click the **Add Interpreter** link or the gear icon, and select **Add Local Interpreter**.
4. Choose **Virtualenv Environment**, select **Existing environment**, and browse to the `python.exe` (Windows) or `python` (macOS/Linux) executable inside your newly created `venv` directory.
5. Click **Apply** and **OK**.

4.2 Step 2: Executing the Notebook

1. In the Project Explorer, right-click your project folder and select **New** → **Jupyter Notebook**. Name it (e.g., `analysis.ipynb`).
2. Open the `.ipynb` file. PyCharm will display its native notebook UI.
3. In the top right corner of the notebook editor, look for the **Jupyter Server** dropdown.

4. Select **Managed Server** (PyCharm will automatically start a local Jupyter server using your configured `venv` interpreter).
5. Type a test command in the first cell, such as:

```
import pandas as pd
print("Jupyter Notebook is running successfully!")
```

6. Click the green **Play** button next to the cell or press **Shift + Enter** to execute the code. The output will appear directly beneath the cell.